

CISCO CERTIFIED CYBEROPS ASSOCIATE (CBROPS 200-201)

Lab 8.2.8: Using Wireshark to Examine Ethernet Frames

Layer 2 Ethernet II Frame Structure, OUI Identification, Broadcast ARP Mechanics & Layer 2 vs. Layer 3 Routing Dynamics

COURSE MODULE	PROTOCOL & FRAMING	TOOLS USED	TOPOLOGY
Module 8: Address Resolution Protocol	Ethernet II (DIX) & ARP (RFC 826)	Wireshark, Mininet, Linux CLI	CyberOps Mininet Topology (H3, R1, S1)

Objectives

- Part 1: Examine the Header Fields in an Ethernet II Frame** — Review standard Ethernet II fields (Preamble, SFD, Destination/Source MAC, EtherType, Data, FCS) and analyze ARP broadcast frame parameters.
- Part 2: Use Wireshark to Capture and Analyze Ethernet Frames** — Launch Mininet network simulation, clear host ARP caches, capture ICMP/ARP traffic for local vs. remote destinations, and evaluate Layer 2 hop-by-hop delivery.
- Reflection & Physical Layer Synchronization** — Understand the role of the 8-byte Preamble / Start Frame Delimiter (SFD) and why it is omitted from software packet captures.

Background / Scenario

When upper-layer protocols (such as IPv4, IPv6, or ARP) transmit data across a local area network, the payload is encapsulated into a Layer 2 Data Link frame. In modern LAN architectures, the dominant framing standard is **Ethernet II**.

Ethernet II frames rely on 48-bit Media Access Control (MAC) addresses for local hardware delivery. Understanding the distinction between local Layer 2 frame forwarding (MAC hop-to-hop) and end-to-end Layer 3 packet routing (IP source-to-destination) is foundational for CyberOps security analysts monitoring network perimeters, analyzing ARP spoofing attacks, and investigating traffic captures.

Part 1: Examine Header Fields in an Ethernet II Frame

Ethernet II Frame Structure & Field Specifications

FIELD NAME	FIELD LENGTH	HEXADECIMAL / BINARY CONTENT	PROTOCOL FUNCTION
Preamble + SFD	8 Bytes (7 + 1)	0xAA...AA + 0xAB (SFD)	Hardware clock synchronization and start of frame indication. (Stripped by NIC).
Destination MAC	6 Bytes (48 bits)	ff:ff:ff:ff:ff:ff (or Unicast)	Hardware address of target node or Layer 2 Broadcast.
Source MAC	6 Bytes (48 bits)	f4:8c:50:62:62:6d	Hardware address of the originating Network Interface Card (NIC).
EtherType (Type)	2 Bytes (16 bits)	0x0806 (ARP) / 0x0800 (IPv4)	Identifies the encapsulated upper-layer network protocol in the payload.
Data Payload	46 – 1500 Bytes	Upper-layer Protocol Data Unit (PDU)	Encapsulated Layer 3 packet or ARP payload (padded to minimum 46 bytes).
FCS (CRC-32)	4 Bytes (32 bits)	Cyclic Redundancy Check	Error-detection checksum computed by sender and verified by receiver NIC.

Part 1 Questions & Detailed Solutions

PART 1.STEP 3 — DESTINATION ADDRESS SIGNIFICANCE

What is significant about the contents of the destination address field?

ANALYST FINDING:

The Destination Address is `ff:ff:ff:ff:ff:ff` (all binary 1s / 48 bits set to 1). This is the **Layer 2 Broadcast MAC Address**.

Significance: When access switches receive a frame addressed to the broadcast MAC, they flood it out all active switchports within the local VLAN/broadcast domain. Every host NIC on the LAN accepts and processes the frame up the protocol stack.

PART 1.STEP 3 — ARP BROADCAST MECHANICS

Why does the PC send out a broadcast ARP prior to sending the first ping request?

PROTOCOL RESOLUTION MECHANICS:

Before the sending host can construct and transmit an IP packet (ICMP Echo Request) over Ethernet, it requires the destination device's physical Layer 2 MAC address to populate the Ethernet II header.

Because the target IP's MAC address is not currently stored in the host's local **ARP Cache table**, the host broadcasts an **ARP Request** ("Who has IP 10.0.0.1? Tell 10.0.0.11") so the owner of that IP address responds with its MAC address.

PART 1.STEP 3 — MAC ADDRESS & OUI ANATOMY

1. What is the MAC address of the source in the first frame?
2. What is the Vendor ID (OUI) of the Source's NIC?
3. What portion of the MAC address is the OUI?
4. What is the Source's NIC serial number?

MAC ADDRESS BREAKDOWN (F4:8C:50:62:62:6D):

- **1. Source MAC Address:** `f4:8c:50:62:62:6d`
- **2. Vendor ID (OUI):** `f4:8c:50` (Registered to **Intel Corporate**).
- **3. OUI Portion:** The **first 3 octets (first 24 bits / 6 hexadecimal characters)** of the 48-bit MAC address comprise the **Organizationally Unique Identifier (OUI)** assigned to the hardware manufacturer by the IEEE.
- **4. NIC Serial Number:** The **last 3 octets (last 24 bits / 6 hexadecimal characters)**: `62:62:6d` represents the unique, vendor-assigned Network Interface Controller serial identifier.

Part 2: Capture and Analyze Ethernet Frames in Mininet

Step 1: Inspect Network Configuration of Host H3

```
[analyst@secOps ~]$ sudo ./lab.support.files/scripts/cyberops_topo.py
mininet> xterm H3

# On Node H3 Terminal:
[root@secOps ~]# ip address
2: H3-eth0: ...
    link/ether 5a:d0:1d:01:9f:be brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.13/24 brd 10.0.0.255 scope global H3-eth0

[root@secOps ~]# netstat -r
Kernel IP routing table
Destination        Gateway           Genmask          Flags   MSS Window  irtt Iface
default            10.0.0.1         0.0.0.0          UG      0 0        0 H3-eth0
10.0.0.0           0.0.0.0         255.255.255.0    U       0 0        0 H3-eth0
```

STEP 1.E — DEFAULT GATEWAY VERIFICATION

What is the IP address of the default gateway for the host H3?

HOST ROUTE SETTING:

The default gateway IP address for host H3 is **10.0.0.1**.

Step 2 to 5: Local Subnet Ping Capture & Dissection (H3 → Gateway 10.0.0.1)

After clearing the ARP cache (`arp -d 10.0.0.11`), Wireshark is started on **H3-eth0** and 5 ICMP packets are sent to **10.0.0.1**. Filtering by **icmp** isolates the ping frames:

STEP 5.C — LOCAL ICMP ECHO REQUEST FRAME DETAILS

1. What is the MAC address of the PC's NIC?
2. What is the default gateway's MAC address?
3. What type of frame is displayed?

DISSECTION FINDINGS (ECHO REQUEST):

- **1. Source MAC (H3 NIC):** **5a:d0:1d:01:9f:be** (Host H3 virtual interface).
- **2. Destination MAC (Default Gateway):** **00:00:00:00:00:01** (or router gateway interface MAC).
- **3. Frame Type:** **Ethernet II** with EtherType field value **0x0800 (IPv4)**.

STEP 5.E & 5.G — ADDRESSING & ECHO REPLY VERIFICATION

1. What is the source and destination IP address in the Echo Request?
2. In the Echo Reply frame, what device and MAC address is displayed as the destination address?

DISSECTION FINDINGS:

- **1. Echo Request IP Header:** Source IP: **10.0.0.13** → Destination IP: **10.0.0.1**.
- **2. Echo Reply Frame Destination:** Device: **Host H3** | Destination MAC: **5a:d0:1d:01:9f:be** (Source and destination MAC addresses reverse during the reply).

Step 6 & 7: Remote Subnet Ping Capture (H3 → Remote Server 172.16.0.40)

From host H3, 5 ICMP echo requests are sent to off-subnet host **172.16.0.40**:

```
[root@secOps analyst]# ping -c 5 172.16.0.40
PING 172.16.0.40 (172.16.0.40) 56(84) bytes of data.
64 bytes from 172.16.0.40: icmp_seq=1 ttl=63 time=0.082 ms
...
```

STEP 7 — REMOTE TRAFFIC ADDRESSING ANALYSIS

1. In the first echo request frame, what are the source and destination MAC addresses?
2. What are the source and destination IP addresses contained in the data field of the frame?

CAPTURED FRAME ATTRIBUTES:

- **Source MAC:** 5a:d0:1d:01:9f:be (Host H3 NIC).
- **Destination MAC:** 00:00:00:00:00:01 (Default Gateway Router LAN Interface).
- **Source IPv4 Address:** 10.0.0.13 (Host H3).
- **Destination IPv4 Address:** 172.16.0.40 (Remote Server).

STEP 7 — CORE LAYER 2 VS. LAYER 3 ROUTING DYNAMIC

Why has the destination IP address changed, while the destination MAC address remained the same?

SOC CORE NETWORKING PRINCIPLE:

1. Layer 3 (IP) Routing Dynamics: IP addresses represent **end-to-end logical addressing**. The destination IP (172.16.0.40) identifies the final remote destination host across the entire internetwork and remains constant from origin to destination.

2. Layer 2 (MAC) Delivery Dynamics: MAC addresses represent **local hop-to-hop physical addressing** and are only valid within the local broadcast domain (LAN).

Conclusion: When host H3 compares the destination IP (172.16.0.40) against its local subnet mask (255.255.255.0), it determines that the target is on a **remote network**. Host H3 cannot directly deliver frames to remote MAC addresses; therefore, it encapsulates the IP packet inside an Ethernet II frame addressed to the **MAC address of its local Default Gateway router** (10.0.0.1). The router decapsulates the Layer 2 header and forwards the packet toward the next hop.

Reflection Question & Physical Layer Synchronization

REFLECTION — THE ETHERNET PREAMBLE & START FRAME DELIMITER

Wireshark does not display the preamble field of a frame header. What does the preamble contain?

PHYSICAL LAYER (PHY) ENGINEERING ANALYSIS:

The Ethernet Preamble field contains an **8-byte (64-bit) synchronization bit stream** structured into two distinct parts:

1. **Preamble (7 Bytes / 56 bits):** An alternating pattern of binary 1s and 0s:

10101010 10101010 10101010 10101010 10101010 10101010 10101010 (or 0xAA in hex).

Purpose: Allows the receiving Physical Layer (PHY) hardware transceiver to synchronize its clock timing with the incoming signal pulses.

2. **Start Frame Delimiter (SFD - 1 Byte / 8 bits):** A specific pattern ending in two consecutive 1s:

10101011 (or 0xAB in hex).

Purpose: Explicitly signals to the receiving hardware that the synchronization sequence is complete and that the actual Destination MAC address begins on the very next bit.

Why Wireshark does not display the Preamble: The Network Interface Card (NIC) hardware PHY controller consumes and strips the 8-byte Preamble/SFD during physical frame synchronization before handing the validated frame to the OS driver and packet capture engine (libpcap).

CyberOps Analyst Layer 2 Security Summary

- **Address Scope** Layer 2 MAC addresses change at every router hop; Layer 3 IP addresses remain constant end-to-end.
- **ARP Cache Trust** ARP is stateless and unauthenticated; hosts accept unsolicited ARP replies, leaving networks vulnerable to **ARP Poisoning / Man-in-the-Middle** attacks.
- **Switch Security** Enterprise networks must enforce **Dynamic ARP Inspection (DAI)** and **Port Security** to validate Layer 2 to Layer 3 bindings against a trusted DHCP Snooping binding database.